



## **CSE461**

Introduction to Robotics Lab

### **Real-World Implication**

Group : 3

Section : 8

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## **1. Introduction:**

Light but time-critical items like medicines, diagnostic samples, patient records and small equipment are handled on a regular basis, in modern hospitals. Conventionally, this type of transportation is reliant on the nursing and support workers that must leave their direct nursing tasks to move things around in the hospital, from the ward to the nurse station, between wards, the pharmacy and lab units. The “Smart Autonomous Hospital Delivery & Safety Monitor Robot” will be tested in a controlled indoor facility layout to replace such a repetitive point to point delivery. The robot uses an Arduino Uno R3 microcontroller, 5 infrared line tracking arrays, coordinate identification via RFID, obstacle avoidance using ultrasonic sensors and a servo-controlled load bay, which opens and closes and reports the load bay status along with its surroundings, by using the 2 temperature and flame sensors.

This report describes the impact such a system can have on the broader "real world" deployment in a hospital. Below are explored the aspects of technical navigation and sensing, but also what the robot should do, how it will relate to persons, institutions, laws and expectations when it enters into the clinical mainstream.

## **2. Purpose:**

This analysis is conducted to observe the effectiveness of the proposed robot in different aspects and render the project feasible and practicable in real world environment beyond the controlled lab environment. In particular, the report seeks to:

- Evaluate the impact the robot has on the day-to-day work and role of the nurse in the hospital.
- Identify potential health risks and benefits with automated delivery of medicines and lab samples.
- Explore safety issues of a robot that will be moving in corridors with patients, staff and visitors.
- Identify legal and regulatory issues a healthcare facing autonomous system would pose.
- Think about the cultural and human characteristics which would either welcome or reject robotic aids for palliative care.

The report is going through these aspects, correlating the project's goals, scope and function (from the initial project proposal) to the tasks that will come with enabling automation with a healthcare environment.

## **3. Societal implications:**

- Minimised non-clinical walkarounds between wards, nurse station, lab and the robot's pharmacy have reduced the repetitive walking activities nurse must undertake with medicines, samples and patient charts thus allowing more time for patient care and reducing the fatigue that often occurs during lengthy nursing shifts.
- Redistribution of work roles: change the work role of junior employees (e.g., from courier to other roles) and/or the decrease or redefinition of the work role of messengers.

This can make things more efficient, but can also pose challenges with regard to labor hours and the introduction of a transition plan rather than transfer to the robot for hospitals.

- An improved delivery consistency: where one can get distracted and/or tired of the manual delivery method, a combined routing and RFID verification of the deliveries reduces the risk of delivering to a ward where the package is not supposed.
- Moderate total cost (~BDT 3,860 in prototype) suggests the possibility of its application in small clinics/straight medical centres/Diagnostic centres that are already in existence in Bangladesh, apart from larger hospitals; and thereby provide a convergence with respect to its operational efficiency.
- Dependence risk: Dependence risk; hospitals need to be reliant on these systems for the deliveries of their critical components – a long-term failure, software bug or maintenance issue could cause a supply chain issue at the ward level.

#### **4. Health implications:**

- The storage compartment is equipped with temperature sensor DHT11 which continuously monitors the air temperature and compares it with a safety temperature of 30°C, thus ensuring the safety of transportation for temperature sensitive medicines and biological samples which cannot be guaranteed in the case of manual transportation, especially in the case of transport in a warm climate.
- Minimal contamination risk: Enclosed, servo-locked space only opened at the known destination, reduces medicines and diagnostic samples being unnecessarily touched by people in the vicinity of the hospital, compared with the risk of open carrying by hand through a crowded corridor.
- Consistent and uninterrupted routing, can speed up how long it takes for a patient's blood sample to arrive in the lab to be tested.
- Potential for over-reliance on manual checks: One DHT11 sensor only measures the temperature of the compartment which may not be accurate enough for the temperature compliance of stored items. Temperature flagging should not replace validations done in the hospital's pharmacist controlled cold chain for sensitive drugs, and is a function of the robot.
- Lower physical walking burden for nurses – particularly on hard shifts – can also help avoid physical walking errors over time, which may lead to harm, and keep nurses healthy and happy at work (indirect wellbeing benefit for staff).

#### **5. Safety implications:**

- Obstacle detection: HC-SR04 ultrasonic sensor mounted in the front keeps obstacles away if they are in the range of 20cm - which is a must for avoiding collision in a hospital corridor where IV stands, wheelchair use, visitors, and movement equipment are to be found.

- Fire and flame response: The IR flame sensor will immediately power off the robot, lock the cargo compartment, and display a red RGB indication and alarm will be constantly activated to warn nearby staff personnel of a fire hazard close to robot's route path – although this is an unlikely event.
- Physical footprint and corridor rules: Robots have a new type of pedestrian traffic, the robot itself, when using obstacle detection. Rules regarding designated lanes, speeds and yielding to stretcher teams and emergency crews would have to be made clear for the robot to not become a nuisance during an emergency.
- Security of payloads – An RFID identification system for cargo compartment cargo locks also provides an added layer of security should there be an attempt to tamper with or steal cargo during transportation; a critical safety aspect regarding controlled substances.
- Realistic deployment: If any one of the several sensors (IR array, RFID reader, ultrasonic sensor) fails, the system would not proceed without knowing the complete set of data, which makes a well-defined fail-safe state a must, such as stopping the system and warning personnel of the failure.
- Battery and electrical safety: The two 18650 Li-ion batteries are a battery pack made up of Lithium-batteries that must be stored and charged as per the safe lithium-battery practices and batteries that are part of a battery pack must be protected from heat or short circuit, if they are continuously running in a hospital environment.

## **6. Legal implications:**

Autonomous service robots in hospitals and other health care facilities should consider legal issues (safety compliance, liability etc.). Medical devices do not have to be registered to be reviewed by hospital administration and local health-facility safety regulations, even if they are not registered as a medical device and are used to transport medicines or diagnostic samples. Clearly, if this is a result of misdelivery, delay or failure in environmental control (e.g. an undetected temperature change) the hospital has the responsibility to establish operational accountability to decide who is responsible for the failure – the hospital, the maintenance team or the system designer. If the system is being utilized for the control of controlled substances, there are additional legal requirements for chain of custody forms and the safe handling of controlled substances, besides the use of the lock and RFID. A further important issue is data security – the RFID routing system could include information on the use of the system but if deliveries were linked to specific patients then there would be serious considerations to meet for data security in relation to patient health data. Finally, the right of way and injury liability arises when using an autonomous mobile system in close proximity to persons and exhaustive test and safety recording of the obstacles must be recorded when implementing the system throughout the hospital.

## **7. Cultural implications:**

It's difficult to add to the healthcare environment, as human interaction is a major priority in healthcare communication. An introduction should be made slowly and the reasons explained to gain patient and staff trust and understanding that the robot is not an alternative to care but an assistive tool. Technology adoption relies heavily on the notion that the technology is a supportive tool to perform "mindless" work, and thereby reduce job loss from the technological advances. Getting employees to accept the technology depends on how it can be explained to them as a helper that will remove repetitive tasks from their work. This system is quite realistic and affordable for applying in the hospitals situated in Bangladesh and similar settings as the price of the project components are quite low and this system can be slowly introduced locally rather than costly importation. Furthermore, a robot person that can be displayed, and, if an alarm is triggered, moved through the wards, can mitigate patients' fear of the security measures implemented in a care facility, and hence their perception of good care. Where onboard LCD is used, all staff and patients should be able to understand what is happening to the system and the alerts without any language barrier, where this interface would be ideal, it should be multilingual with the local language and English.

## **8. Conclusion:**

The Smart Autonomous Hospital Delivery & Safety Monitor Robot can also reduce the burden on hospital staff by eliminating the need to manually open or close the door with a servo-locked cargo door, removing the need for a human hand; use IR line tracking to track the direction of delivery, ensuring the correct delivery direction; use two hazard sensors to verify the coordinates with cargo and avoid collisions; eliminate the need for manpower to check coordinates by using RFID coordinate verification; and provide the ability to perform ultrasonic obstacle avoidance. At the same time, the move from a laboratory prototype to a working healthcare environment will be more than engineering change - it's responsibilities in social adjustment, with staff roles, health security for vulnerable commodities, physical safety of the corridors, legal responsibilities for provision and outcomes, and cultural aspects of introducing automation into a care environment.

In general, the main functions of the project directly cater to these real-life demands, especially the collision avoidance behavior, temperature and flame detection and secure RFID-gated compartment. But to deploy successfully, as well as at the circuit level, it would need supporting measures that are not being discussed here, including formal safety certification, a proper liability programme, training and a slow cultural rehabilitation. The supporting measures that were proposed make the suggested robot a viable and affordable solution to a safer, more efficient, and more consistent way of handling materials in a hospital environment.